

Problems

Section 15 — Batteries

Problem 15.1

A battery with a capacity of 1800 mAh is to be used in a device that draws 120 mA continuously. Ignoring possible loss in capacity as a result of load current magnitude, how long should the battery be able to deliver power?

Problem 15.2

Determine the discharge time and average current output of a battery with a capacity rating of 1000 mAh if it is discharged at 1 C.

How long would it take to discharge at 5 C, 2 C, 0.5 C, 0.2 C and 0.05 C?

Section 16 — Resistors

Problem 16.1

If a $1000\ \Omega$ resistor is connected in parallel with one of $3000\ \Omega$, what is the total (equivalent) resistance?

Problem 16.2

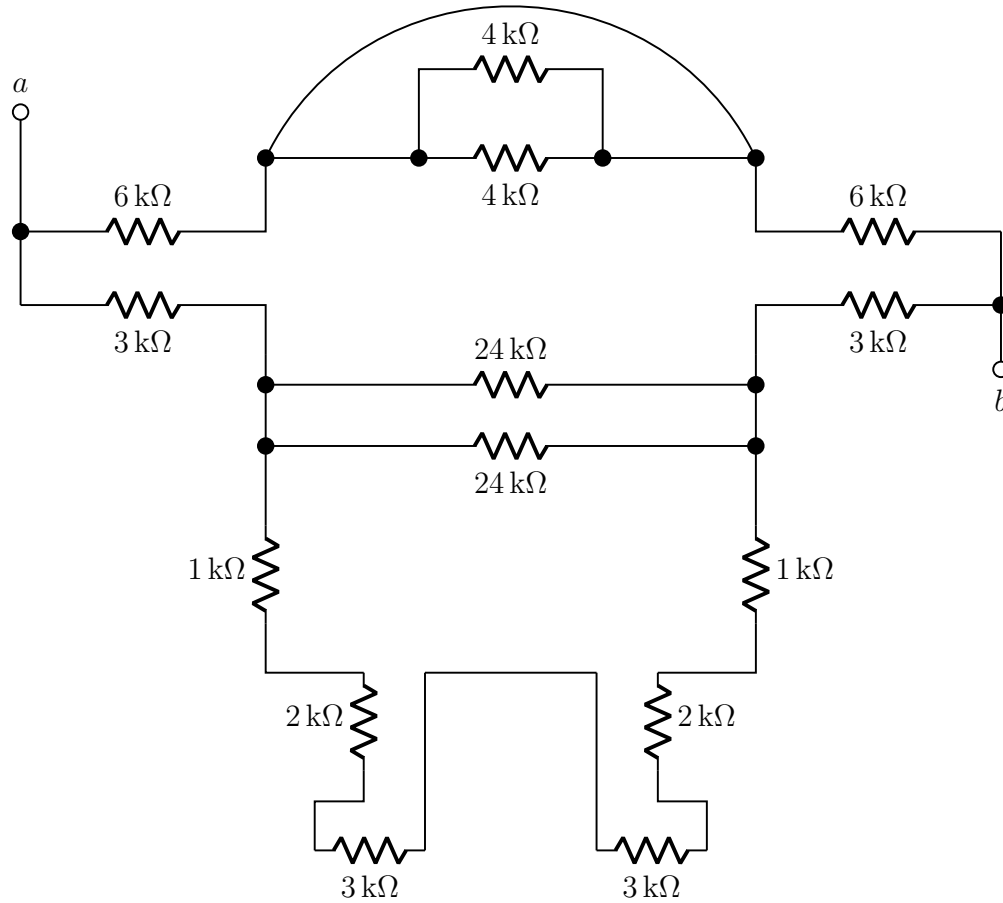
If a $1.0\ \text{k}\Omega$ resistor is placed in series with a $2.0\ \text{k}\Omega$ resistor, what is the total resistance?

If the two resistors are then attached to an input voltage of $V_{in} = 9\ \text{V}$, find:

1. The total current flow and the current through each resistor.
2. The voltage drop across each resistor.

Problem 16.3

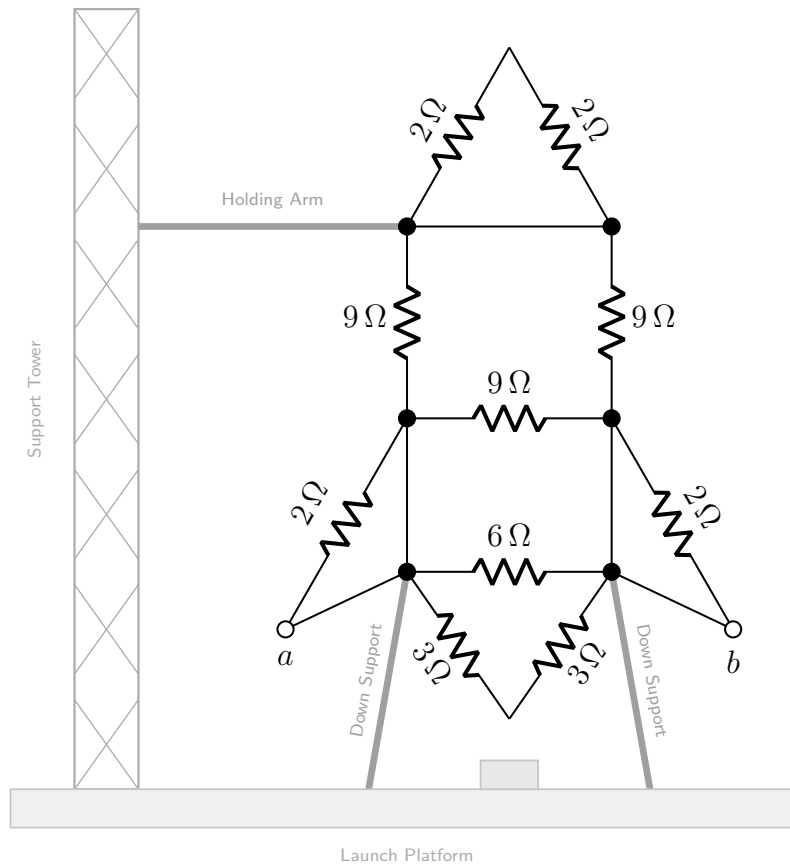
Find the equivalent resistance between terminals a and b of the resistor network shown below.



Problem 16.4

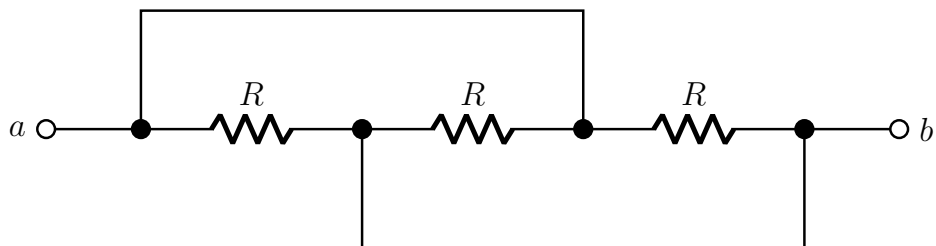
Find the equivalent resistance between terminals a and b of the resistor network shown below. All resistances are given in ohms (Ω).

The support tower, holding arm, down supports and launch platform (drawn in grey) are mechanical structures only — they are **not** electrical connections.



Problem 16.5

In the circuit below, all three resistors have the same resistance R . Find the equivalent resistance between the terminals a and b . What is the configuration of the three resistors — series, parallel, or neither?

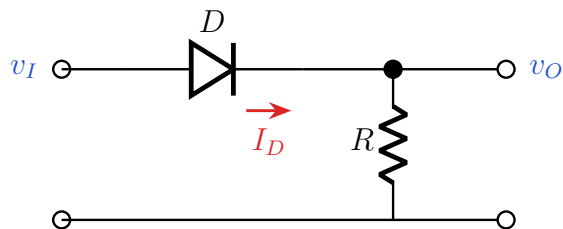


Section 20 — Diodes

In every problem, validate your assumptions after solving.

Problem 20.1

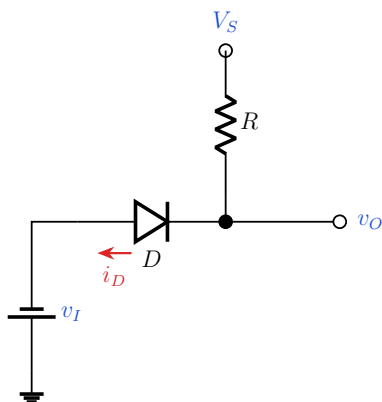
For the circuit below, if $v_I = 5\text{ V}$ and $V_{DO} = 0.7\text{ V}$, find v_O .



Problem 20.2

For the circuit below, find v_O in each of the following cases. Take $V_{DO} = 0.7$ V throughout.

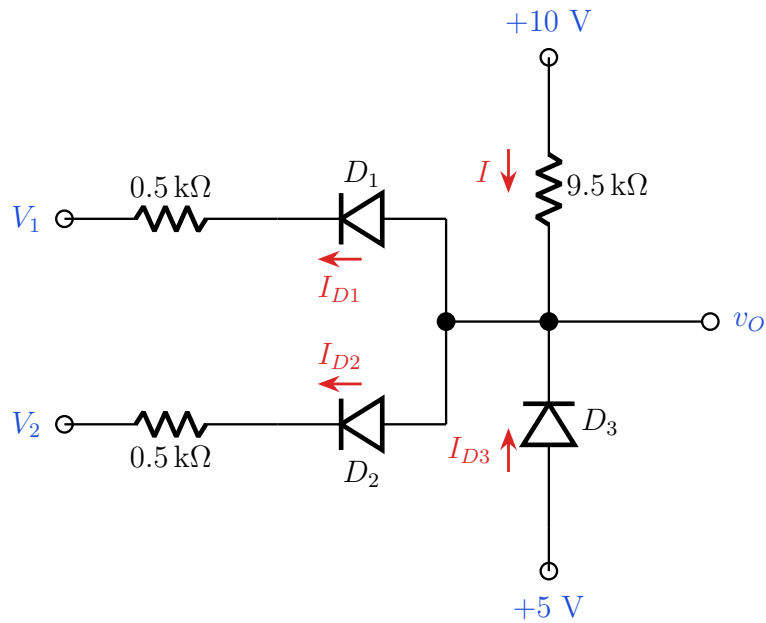
1. $v_I = 5$ V, $V_S = 10$ V
2. $v_I = 11$ V, $V_S = 10$ V
3. $v_I = 0$ V, $V_S = 10$ V



Problem 20.3

Analyse the circuit below to find I_{D1} , I_{D2} , I_{D3} and I for each of the following input combinations. Use the **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50$ Ω .

1. $V_1 = 0$ V, $V_2 = 2$ V
2. $V_1 = V_2 = 0$ V
3. $V_1 = V_2 = 5$ V
4. $V_1 = 5$ V, $V_2 = 0$ V



Problem 20.4

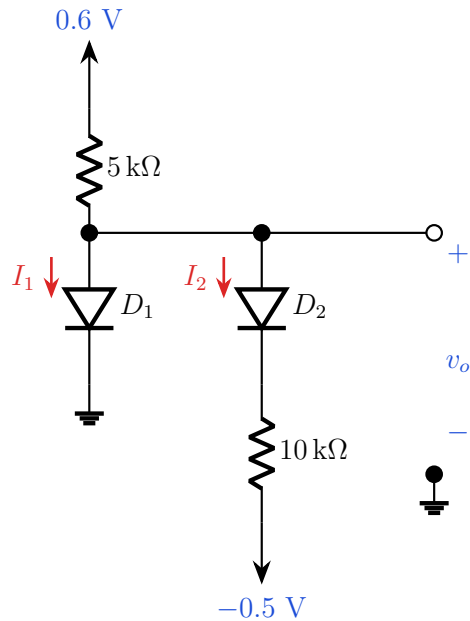
For the same circuit as Problem 20.3, replace the $9.5 \text{ k}\Omega$ pull-up resistor with a resistor of value X (in $\text{k}\Omega$) and take $V_{DO} = 0.6 \text{ V}$.

1. Repeat the analysis of Problem 20.3 for $X = 9.5$.
2. For what value of X will D_3 be in cut-off?

Problem 20.5

Analyse the circuit below to find I_1 , I_2 and v_o under each of the following diode models.

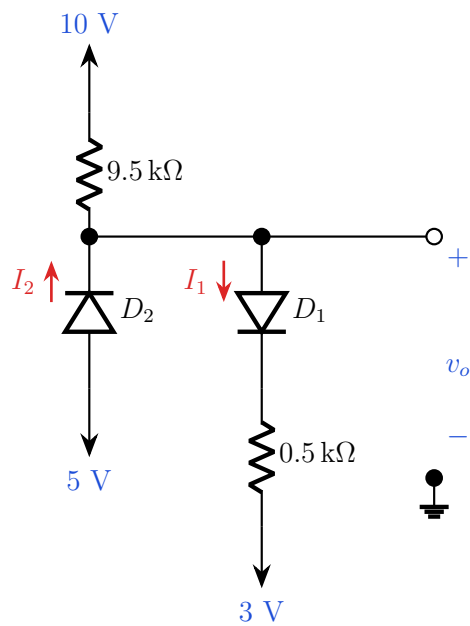
1. All diodes assumed **ideal**.
2. **CVD model** with $V_{D1} = 0.5 \text{ V}$ and $V_{D2} = 0.7 \text{ V}$.
3. **CVD+R model** with $V_{DO} = 0.6 \text{ V}$ and $r_d = 50 \Omega$.



Problem 20.6

Analyse the circuit below to find I_1 , I_2 and v_o under each of the following diode models.

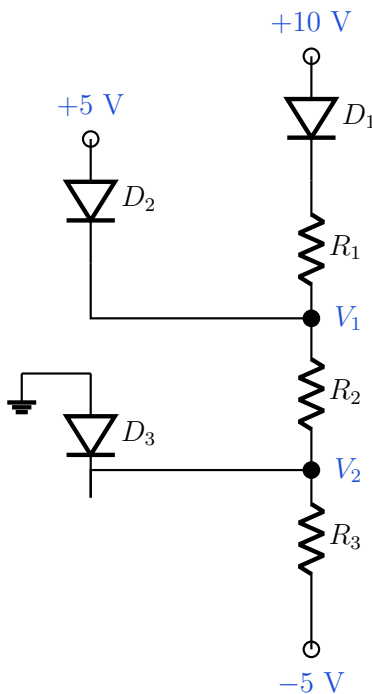
1. All diodes assumed **ideal**.
2. **CVD model** with $V_{DO} = 0.6$ V.
3. **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50$ Ω .



Problem 20.7

For the circuit below, take $V_{DO} = 0.6$ V.

1. Determine R_1 , R_2 and R_3 such that $I_{D1} = 0.2$ mA, $I_{D2} = 0.3$ mA and $I_{D3} = 0.5$ mA.
2. Find V_1 , V_2 and each diode current for $R_1 = 10$ k Ω , $R_2 = 4$ k Ω and $R_3 = 2.2$ k Ω .



Problem 20.8

For the same circuit as Problem 20.7 with $V_{DO} = 0.6$ V, find V_1 , V_2 and each diode current for:

1. $R_1 = 3$ k Ω , $R_2 = 6$ k Ω , $R_3 = 2.5$ k Ω
2. $R_1 = 6$ k Ω , $R_2 = 3$ k Ω , $R_3 = 6$ k Ω

Section 21 — Rectifiers

Validate your assumptions where a diode state must be chosen.

Problem 21.1

A voltage waveform $v_i = 10 \sin(100\pi t)$ V is input to a full-wave rectifier with a load resistance of $R = 50$ k Ω . Silicon diodes are used in this circuit, for which the forward drop is $V_{D0} = 0.7$ V.

1. **Show** the circuit of the rectifier. **Label** the input and output voltages properly. [2]
2. **Calculate** the DC value of the output voltage. [1]
3. **Contrast** the value found in part (b) with that when a 5 μ F capacitor is connected in parallel with the load. [2]

4. **Identify** the two diodes that will be ON in the positive half cycle. [1]

Now the two diodes from part (d) are replaced with germanium diodes ($V_{D_0} = 0.2$ V).

5. **Explain** the change in the voltage transfer characteristics and output voltage waveform of the circuit. Hence, calculate the peak of the output voltage in this case. [3+1]

Problem 21.2

A voltage waveform $v_i = 5 \sin(200\pi t)$ V is fed into a full-wave rectifier with a load resistor $R = 5$ k Ω . Silicon diodes are used in this circuit, where $V_{D_0} = 0.6$ V.

1. **Draw** the rectifier circuit. **Label** the input and output voltages properly. Briefly **explain** the application of the circuit. [1+1+1]
2. **Calculate** the DC value of the output voltage, V_{dc} , and the output frequency, f_o . [1+1]
3. **Draw** the Voltage Transfer Characteristics (VTC) of the full-wave rectifier and **label** it properly. [2]
4. Now you have to connect a capacitor in parallel with the load resistor. You have two capacitors of 5 μ F and 1 μ F at your disposal. Which capacitor will you use? **Explain** briefly with necessary calculations. [3]
5. **[Bonus]** A different input waveform is fed into the full-wave rectifier. The new peak-to-peak ripple voltage is 50% of the previous one calculated in part (d) with the 5 μ F capacitor. The new output frequency is 300 Hz. **Determine** the equation of the input waveform. [2]

Problem 21.3

A voltage waveform $v_i = 10 \sin(1000\pi t)$ V is fed into a full-wave (FW) rectifier with a load resistance $R = 10$ k Ω . A capacitor is also connected in parallel with the load to reduce the fluctuation of the output voltage. It produces a peak-to-peak ripple voltage which is **3%** of the peak output voltage. The diodes have a forward voltage drop of $V_{D_0} = 0.8$ V.

1. **Deduce** the peak output voltage, V_p , and the peak-to-peak ripple voltage, $v_{r(p-p)}$. [2]
2. **Calculate** the average (DC) value of the output voltage. [1]
3. **Estimate** the value of the capacitor from the given data. [2]

Now the input voltage v_i is changed to the one shown in Figure (ii) of the original handout, and the **capacitor is removed**.

4. **Show** the output waveform and **indicate** the voltage levels properly. [2]

Problem 21.4

A voltage waveform $v_i = 8 \sin(2000\pi t)$ V is input to a full-wave rectifier. A resistance of $R = 50$ k Ω is connected at the load. Assume that the diodes used in the circuit have a forward drop of 0.8 V.

1. **Draw** the circuit of the full-wave rectifier. Label the input and output voltages properly. [1]
2. **Draw** the waveforms of the input and output voltages. What are the peak values of input and output? Show them in the graph. [1+1]

3. **Find** the average voltage measured at the output.

[1]

Solutions

Section 15 — Batteries

Solution 15.1

A battery with a capacity of 1800 mAh is to be used in a device that draws 120 mA continuously. Ignoring possible loss in capacity as a result of load current magnitude, how long should the battery be able to deliver power?

The time of operation before the battery is fully drained is the battery capacity divided by the current drawn.

$$t = \frac{\text{Battery Capacity}}{\text{Current}} = \frac{1800 \text{ mAh}}{120 \text{ mA}} = 15 \text{ h}$$

Note: In reality, the manufacturer's data sheets and discharge graphs (voltage as a function of time and of load current) must be consulted to get an accurate determination of the actual discharge time. As the load current increases, there is an apparent loss in battery capacity caused by internal resistance.

Solution 15.2

Determine the discharge time and average current output of a battery with a capacity rating of 1000 mAh if it is discharged at 1 C.

How long would it take to discharge at 5 C, 2 C, 0.5 C, 0.2 C and 0.05 C?

At a discharge rate of 1 C, a battery delivers a current equal to its rated capacity for one hour. The discharge time is therefore

$$t = \frac{1 \text{ hC}}{\text{C rating}}$$

At 1 C, the battery is attached to a load drawing 1000 mA (rated capacity per hour), so

$$t = \frac{1 \text{ hC}}{1 \text{ C}} = 1 \text{ h}$$

Applying the same relation for each of the remaining C ratings:

C Rating	Average Current Output	Discharge Time
5 C	5000 mA	0.2 h
2 C	2000 mA	0.5 h
1 C	1000 mA	1 h
0.5 C	500 mA	2 h
0.2 C	200 mA	5 h
0.05 C	50 mA	20 h

Note: These values are estimates. When load currents increase (especially when C values get large), the capacity level drops below nominal values due to non-ideal internal characteristics such as internal resistance, and must be determined using the manufacturer's discharge curves and Peukert's equation.

Section 16 — Resistors

Solution 16.1

If a 1000 Ω resistor is connected in parallel with one of 3000 Ω , what is the total (equivalent) resistance?

For two resistors in parallel, the equivalent resistance follows from

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \implies R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

Substituting the given values,

$$R_{eq} = \frac{1000 \Omega \times 3000 \Omega}{1000 \Omega + 3000 \Omega} = \frac{3000000 \Omega^2}{4000 \Omega} = 750 \Omega$$

Note that the equivalent resistance (750 Ω) is lower than even the smaller of the two individual resistances (1000 Ω), as expected for a parallel combination.

Solution 16.2

If a 1.0 k Ω resistor is placed in series with a 2.0 k Ω resistor, what is the total resistance?

If the two resistors are then attached to an input voltage of $V_{in} = 9$ V, find:

- 1. The total current flow and the current through each resistor.*
- 2. The voltage drop across each resistor.*

For resistors in series, the equivalent resistance is the sum of the individual resistances.

$$R_{eq} = R_1 + R_2 = 1000 \Omega + 2000 \Omega = 3000 \Omega = 3 \text{ k}\Omega$$

1. Current flow

The current through resistors in series is the same everywhere in the branch, so by Ohm's law,

$$I_{tot} = \frac{V_{in}}{R_{eq}} = \frac{9 \text{ V}}{3000 \Omega} = 0.003 \text{ A} = 3 \text{ mA}$$

$$I_1 = I_2 = 3 \text{ mA}$$

2. Voltage drops

Applying Ohm's law across each resistor,

$$V_1 = I_1 \times R_1 = 0.003 \text{ A} \times 1000 \text{ } \Omega = 3 \text{ V}$$

$$V_2 = I_2 \times R_2 = 0.003 \text{ A} \times 2000 \text{ } \Omega = 6 \text{ V}$$

The same result follows directly from the voltage divider equations:

$$V_1 = \frac{R_1}{R_1 + R_2} \cdot V_{in} = \frac{1000 \text{ } \Omega}{3000 \text{ } \Omega} \times 9 \text{ V} = 3 \text{ V}$$

$$V_2 = \frac{R_2}{R_1 + R_2} \cdot V_{in} = \frac{2000 \text{ } \Omega}{3000 \text{ } \Omega} \times 9 \text{ V} = 6 \text{ V}$$

Notice that the voltage drop across each resistor is directly proportional to its resistance — the $2000 \text{ } \Omega$ resistor is twice as large as the $1000 \text{ } \Omega$ resistor, and its voltage drop is twice as large.

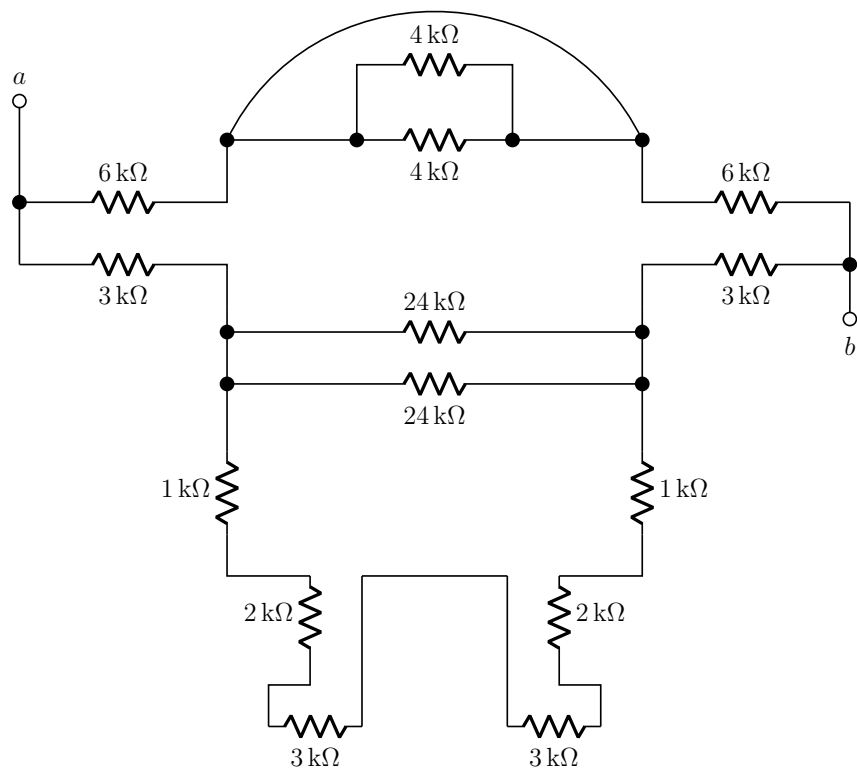
Also note that the voltage drops add up to the supply voltage:

$$+9 \text{ V} - 3 \text{ V} - 6 \text{ V} = 0 \text{ V}$$

That is, the sum of the voltage changes around a closed path is zero. This is **Kirchhoff's voltage law**.

Solution 16.3

Find the equivalent resistance between terminals a and b of the resistor network shown below.



Step 1 — The head is short circuited

The curved wire arcing over the head connects the top-left node directly to the top-right node. Since this path is a plain wire, both $4\text{ k}\Omega$ resistors have their two terminals connected together, so they are **short circuited** and carry no useful current. The head therefore contributes $0\text{ }\Omega$ between the two shoulder nodes.

Step 2 — The upper (head) branch

Going from a through the head to b :

$$R_{head} = 6\text{ k}\Omega + 0\text{ }\Omega + 6\text{ k}\Omega = 12\text{ k}\Omega$$

Step 3 — The torso

The two $24\text{ k}\Omega$ resistors are connected between the same two nodes, so they are in parallel:

$$R_{torso} = \frac{24\text{ k}\Omega \times 24\text{ k}\Omega}{24\text{ k}\Omega + 24\text{ k}\Omega} = 12\text{ k}\Omega$$

Step 4 — The legs

Starting at the left torso node, the legs form a single series path down the left leg, across the bottom, and back up the right leg:

$$R_{legs} = 1 + 2 + 3 + 3 + 2 + 1 = 12\text{ k}\Omega$$

Step 5 — Torso in parallel with legs

The torso and the legs both connect the same two nodes (left torso node and right torso node), so they are in parallel:

$$R_{body} = \frac{12 \text{ k}\Omega \times 12 \text{ k}\Omega}{12 \text{ k}\Omega + 12 \text{ k}\Omega} = 6 \text{ k}\Omega$$

Step 6 — The lower (body) branch

Going from a through the body to b :

$$R_{lower} = 3 \text{ k}\Omega + 6 \text{ k}\Omega + 3 \text{ k}\Omega = 12 \text{ k}\Omega$$

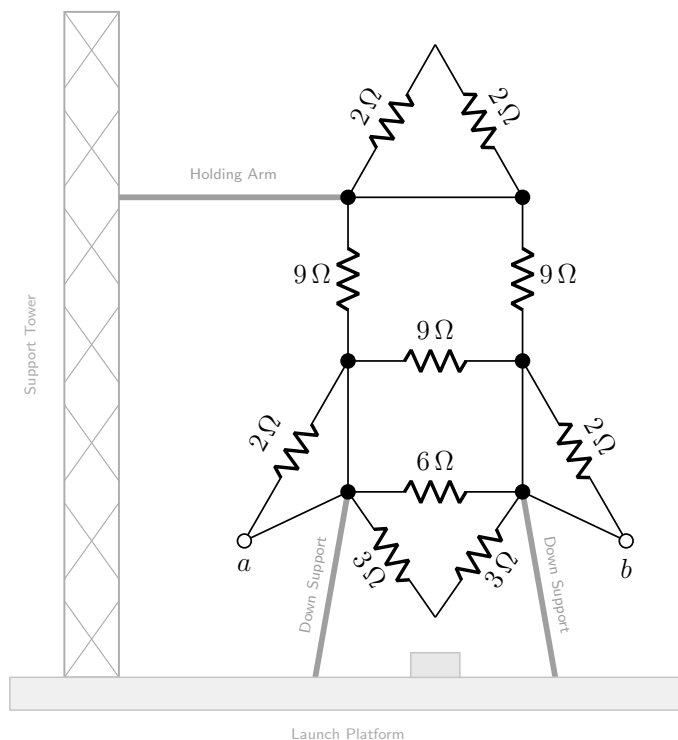
Step 7 — Combine the two branches

The head branch and the body branch both connect terminal a to terminal b , so they are in parallel:

$$R_{ab} = \frac{12 \text{ k}\Omega \times 12 \text{ k}\Omega}{12 \text{ k}\Omega + 12 \text{ k}\Omega} = \boxed{6 \text{ k}\Omega}$$

Solution 16.4

Find the equivalent resistance between terminals a and b of the resistor network shown below. All resistances are given in ohms (Ω).



Step 1 — The nose cone is short circuited

The horizontal wire joining the two top nodes (where the holding arm attaches) connects both ends of the nose cone together. The two $2 \text{ }\Omega$ resistors in the nose are therefore **short circuited** and contribute nothing.

Label the two middle nodes — where the horizontal $9\ \Omega$ resistor connects — as M_L and M_R . Note that the plain vertical wires make the lower body nodes the *same* nodes as M_L and M_R .

Step 2 — The terminal leads are also short circuited

Terminal a reaches the left side of the network by two separate paths: through the $2\ \Omega$ lead resistor up to M_L , and through a plain wire down to the lower left node. Since the lower left node and M_L are the *same* node, that plain wire places a short across the $2\ \Omega$ lead resistor, so it too carries no useful current.

The same argument applies on the right. Therefore

$$a \equiv M_L \quad \text{and} \quad b \equiv M_R$$

and the problem reduces to finding the resistance between M_L and M_R .

Step 3 — The bottom section (fins)

The two $3\ \Omega$ fins are in series with each other:

$$3\ \Omega + 3\ \Omega = 6\ \Omega$$

This series pair is in parallel with the horizontal $6\ \Omega$ resistor:

$$R_{bottom} = \frac{6\ \Omega \times 6\ \Omega}{6\ \Omega + 6\ \Omega} = 3\ \Omega$$

Step 4 — The top section

Going from M_L up through the $9\ \Omega$, across the shorting wire, and down the other $9\ \Omega$ to M_R :

$$R_{top} = 9\ \Omega + 9\ \Omega = 18\ \Omega$$

Step 5 — Combine everything between M_L and M_R

Three paths connect M_L to M_R — the top section, the middle $9\ \Omega$ resistor, and the bottom section — so all three are in parallel:

$$\frac{1}{R_{ab}} = \frac{1}{18\ \Omega} + \frac{1}{9\ \Omega} + \frac{1}{3\ \Omega} = \frac{1 + 2 + 6}{18\ \Omega} = \frac{9}{18\ \Omega} = \frac{1}{2\ \Omega}$$

$$R_{ab} = \boxed{2\ \Omega}$$